CASE

Tyler, 16 · woke at 5 AM with **severe right scrotal pain** · nausea, vomiting · right testicle **high-riding and horizontal** · **absent cremasteric reflex** · color Doppler: absent blood flow · **testicular torsion**

BIO 004 · WEEK 6 · CASE DEEP DIVE

The Male Reproductive System

A teenager woken by sudden testicular pain. A high-riding testicle. A six-hour clock that started the moment the cord twisted.

PATIENT CASE

Tyler is 16. He was woken at 5 AM by **severe pain in his right testicle** and lower abdomen. He vomited. His mother brought him to the ED at 6:30 AM.

On exam the right testicle is **high-riding compared to the left, oriented horizontally rather than vertically, and exquisitely tender**. The **cremasteric reflex is absent on the right** (present on the left). Color Doppler ultrasound shows **absent blood flow to the right testicle**. The diagnosis is **testicular torsion**. Urology is paged immediately; the testicle is salvageable for about 6 hours from the start of pain.

Tyler's case rests on the anatomy of the spermatic cord — how the testicle is suspended, what attaches it to the scrotum, and why it can rotate and cut off its own blood supply.

> **GUIDING QUESTION**
>
> How is the male reproductive system organized, what travels through the spermatic cord, and what anatomic variation lets a testicle twist itself off its blood supply?

PREWORK Companion to your Week 6 Male Reproductive prework page.

1 PLAN

Where sperm are made, matured, and delivered

The male reproductive system has two functional jobs: produce sperm and deliver them. Anatomy follows: **testes** (produce sperm and testosterone), **duct system** (epididymis → ductus deferens → ejaculatory duct → urethra), **accessory glands** (seminal vesicles, prostate, bulbourethral), **penis** (delivery).

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2 TESTES

Suspended in the scrotum for a reason

The testes are paired ovoid organs in the **scrotum**, which sits outside the body to keep them ~2°C below core (sperm production requires lower temperature). Each testis is wrapped in a tough fibrous capsule (tunica albuginea) sending in septa that divide it into ~250 lobules.

- **Seminiferous tubules**
  1-4 tightly coiled tubules per lobule. *Site of spermatogenesis.* Lined by germ cells (sperm precursors) and *Sertoli cells* (nurse cells that support spermatogenesis and form the blood-testis barrier).

- **Interstitial (Leydig) cells**
  Between tubules. Produce **testosterone** in response to LH from the anterior pituitary.

- **Rete testis & efferent ductules**
  Sperm collect here, then move into the epididymis.

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3 DUCT SYSTEM

Where sperm mature and travel

Epididymis
Long coiled tube on the back of the testis. Sperm mature here (~20 days) and gain motility.

Ductus (vas) deferens
Muscular tube from epididymis up through the inguinal canal, into the pelvis, behind the bladder. *Cut and tied in vasectomy.*

Ejaculatory duct
Formed where the ductus deferens meets the seminal vesicle's duct. Empties into the prostatic urethra.

Urethra
Three parts: prostatic, membranous, spongy. Carries both urine and semen (not simultaneously — sphincters coordinate).

4 SPERMATIC CORD

What goes through the inguinal canal

The **spermatic cord** suspends the testis and carries everything that enters or leaves it through the inguinal canal:

- **Ductus deferens**
- **Testicular artery** (from abdominal aorta)
- **Pampiniform venous plexus** (drains to testicular vein → IVC on right, left renal vein on left)
- **Lymphatics** (drain to para-aortic nodes, not inguinal — why testicular cancer doesn't typically present with inguinal nodes)
- **Autonomic nerves**
- Surrounded by the *cremaster muscle* (controls scrotal height for temperature regulation; absent contraction = absent cremasteric reflex)

> **Why torsion happens.** The testicle is normally tethered posteriorly by the tunica vaginalis. In some boys (~10% — the "bell clapper" deformity), the testis is suspended fully within the tunica vaginalis with no posterior fixation. It can rotate freely on the spermatic cord. When it twists (often during sleep or activity), it occludes the testicular artery first. Pain is immediate and severe. Without restoration of flow within ~6 hours, the testicle becomes necrotic.

5 ACCESSORY GLANDS

Contribute the fluid component of semen

Seminal vesicles (paired)
Behind bladder. Contribute ~60% of semen volume: fructose (energy for sperm), prostaglandins, clotting factors.

Prostate
Surrounds the urethra below the bladder. Contributes ~30% of semen: milky fluid with PSA (liquefies clotted semen). Enlarges with age (BPH).

Bulbourethral (Cowper's) glands
Small pea-sized glands inferior to prostate. Pre-ejaculatory mucus that lubricates and neutralizes acidic urine in the urethra.

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6 PENIS

Erection mechanics in three columns

Three columns of erectile tissue: paired **corpora cavernosa** dorsally, single **corpus spongiosum** ventrally (which surrounds the urethra and forms the glans distally). Erection: parasympathetic input → NO release → arteriolar dilation → cavernous sinuses fill with blood → swelling compresses venous outflow → rigid erection. Ejaculation: sympathetic.

✎ **DRAW ZONE 1** -- *Sketch the male reproductive system in sagittal section. Label testes, epididymis, ductus deferens (through inguinal canal), seminal vesicle, prostate, ejaculatory duct, urethra, penis.*

Trace the path of sperm from seminiferous tubule to outside.

✎ **DRAW ZONE 2** *-- Diagram a cross-section of the spermatic cord with all contents labeled. Then show how the testicle can twist on the cord (bell-clapper) and occlude blood flow.*

Star the testicular artery and pampiniform plexus. Show the twist and the cutoff perfusion below it.

RETURNING TO TYLER

A clock that started at 5 AM

Every finding on Tyler's exam points at the spermatic cord. Torsion is one of the few true urologic emergencies; anatomy determines whether his testicle survives.

"*Sudden severe pain on awakening*"

Cremasteric muscle relaxation during sleep can let the testis rotate. Patients classically present in the early morning hours.

"*High-riding, horizontal testicle*"

Twisting shortens the cord and lifts the testis. Loss of normal vertical orientation is a classic sign.

"*Absent cremasteric reflex*"

The reflex (stroking the inner thigh elevates the testis on the same side) requires intact innervation and an unaffected cord. Loss is sensitive for torsion.

"*Absent flow on Doppler*"

Confirms vascular occlusion. Some institutions skip imaging if exam is classic and go straight to OR.

"*Why time matters*"

Testicular salvage rates: ~90% within 6 hours, ~50% by 12 hours, <10% after 24 hours. Tyler is at 1.5 hours of pain — well within the window.

"*Treatment*"

Surgical detorsion + bilateral orchiopexy (suturing both testicles to the scrotal wall to prevent recurrence on either side, since bell-clapper deformity is often bilateral). Manual detorsion can be attempted at bedside but doesn't replace surgery.

Cord, artery, salvage window. Tyler's anatomy gave the testicle the freedom to twist; the clock decides whether anatomy gives it back.